FN4 Worked Examples: Transformations of Functions

Recording companion for the FN4 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Where does each key point go? Read $g(x) = -f(x + b) + c$ from the inside out. The number inside the parentheses moves the graph sideways, and it moves the opposite way from its sign: $x + 3$ moves left 3. The minus in front flips the graph across the x-axis. The number added at the end moves it up or down. To sketch g, move the endpoints and the turning point one step at a time, then connect them with the same shape.

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Example 1: Read 1

$$g(x)=f(x-2)+3$$

Question

How is the graph of g related to the graph of f?

- **A.** Shift left 2, then up 3
- **B.** Shift right 3, then up 2
- **C.** Shift right 2, then up 3
- **D.** Shift left 2, then down 3

Answer and reasoning

Inside the parentheses, $x-2$ shifts the graph right 2: the input has to be 2 bigger to get the same output, so every point slides 2 to the right. Outside, $+3$ shifts it up 3, the same way as the sign. Reading the inside sign at face value gives "left 2", and the 2 and 3 are not interchangeable. **Shift right 2, then up 3.**

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Example 2: Read 2

$$g(x)=-f(x+4)-1$$

Question

How is the graph of g related to the graph of f, in order?

A. Shift left 4, reflect across the x-axis, then shift down 1
B. Shift right 4, reflect across the x-axis, then shift down 1
C. Shift left 4, reflect across the y-axis, then shift down 1
D. Shift left 4, shift down 1, then reflect across the x-axis

Answer and reasoning

$x+4$ shifts left 4. The minus in front of f flips every y-value, a reflection across the x-axis. Then -1 shifts down 1. Order matters: shifting down first and then reflecting would end up 1 above the reflected graph, which is $-(f(x+4)-1)=-f(x+4)+1$. A reflection across the y-axis would need the minus inside, $f(-x)$. **Left 4, reflect across the x-axis, then down 1.**

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Example 3: Read 3

$$g(x)=-f(x-1)+5$$

Question

How is the graph of g related to the graph of f, in order?

- **A.** Shift right 1, shift up 5, then reflect across the x-axis
- **B.** Shift left 1, reflect across the x-axis, then shift up 5
- **C.** Shift right 1, reflect across the y-axis, then shift up 5
- **D.** Shift right 1, reflect across the x-axis, then shift up 5

Answer and reasoning

$x-1$ shifts right 1. The minus in front reflects across the x-axis. Then $+5$ shifts up 5, and it comes last: if you shifted up 5 first and then reflected, the graph would end up 5 below the reflected graph instead of 5 above it. $x-1$ goes right, not left, and a y-axis reflection would need $f(-x)$. **Right 1, reflect across the x-axis, then up 5.**

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Example 4: Read 4

$$g(x)=f(x+5)-4$$

Question

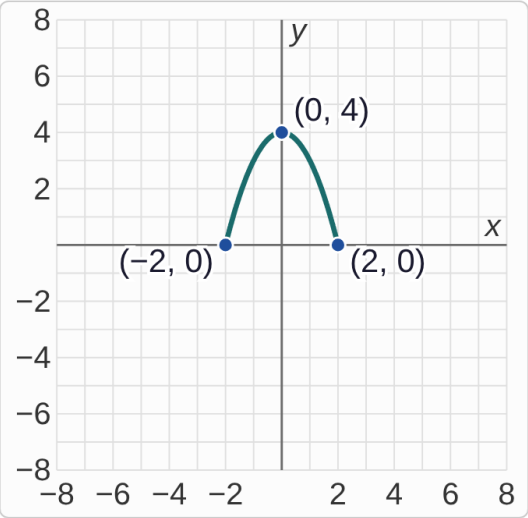
The point (0, 4) is on the graph of f. Which point is on the graph of g?

A. (5, 0)
B. (−5, 0)
C. (−5, 8)
D. (5, 8)

Answer and reasoning

$x+5$ moves every point left 5, so $x=0$ becomes $x=-5$. Then -4 moves it down 4, so $y=4$ becomes $y=0$. Check: $g(-5)=f(-5+5)-4=f(0)-4=4-4=0$. The other points move right or up instead. ***(−5,0)*.**

Example 5: Track 1



Plotted data: The graph of f on a grid from −8 to 8 in both directions. It is an arch of the parabola y = 4 − x squared, drawn only from x = −2 to x = 2. It starts at the point (−2, 0), rises to its top point (0, 4), and comes back down to (2, 0). All three points are labeled.

$$g(x)=f(x-3)+1$$

Question

Where does the top point (0, 4) of f end up on the graph of g?

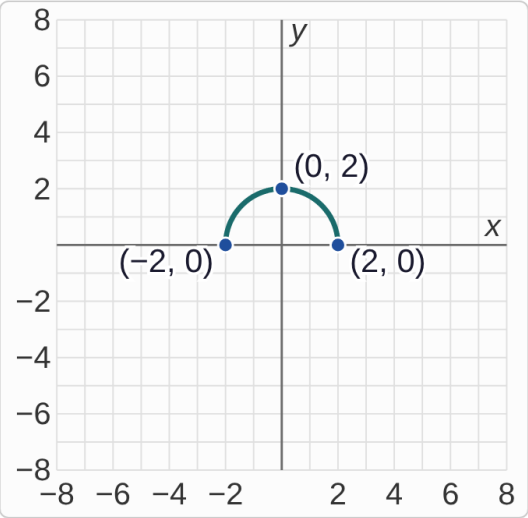
- **A.** (3, 5)
- **B.** (−3, 5)
- **C.** (3, 3)
- **D.** (−3, 3)

Answer and reasoning

x−3 moves the point right 3, to *(3,4)*. Then *+1* moves it up 1, to *(3,5)*. The endpoints go along: *(−2,0)* goes to *(1,1)* and *(2,0)* goes to *(5,1)*. Sketch those three points and draw the same arch through them. ***(3,5)*.**

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Example 6: Track 2



Plotted data: The graph of f on a grid from −8 to 8 in both directions. It is the top half of a circle of radius 2 centered at the origin. It starts at (−2, 0), rises to its top point (0, 2), and comes back down to (2, 0). All three points are labeled.

$$g(x)=-f(x+1)+3$$

Question

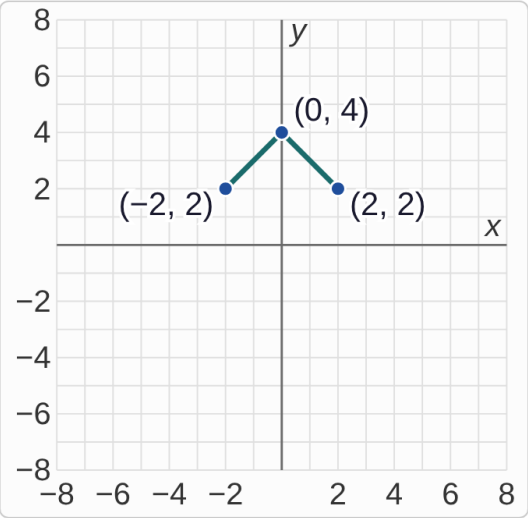
Where do the three labeled points go on the graph of g?

- **A.** (−1, 3), (1, 1), (3, 3)
- **B.** (−3, 3), (−1, 1), (1, 3)
- **C.** (−3, 3), (−1, 5), (1, 3)
- **D.** (−3, −3), (−1, −5), (1, −3)

Answer and reasoning

Step by step: left 1 gives *(−3,0),(−1,2),(1,0)*. Reflecting across the *x*-axis changes the signs of the *y*'s: *(−3,0),(−1,−2),(1,0)*. Up 3 gives *(−3,3),(−1,1),(1,3)*: the semicircle is now upside down, a bowl whose bottom is at *(−1,1)*. The list starting at *(−1,3)* goes right, the one with *(−1,5)* skips the reflection, and the all-negative list shifts up before reflecting. ***(−3,3), (−1,1), (1,3)*.**

Example 7: Track 3



Plotted data: The graph of f on a grid from −8 to 8 in both directions. It is a tent made of two straight segments: from (−2, 2) up to the peak (0, 4), then down to (2, 2). All three points are labeled.

$$g(x)=f(x+2)-5$$

Question

Where do the three labeled points go on the graph of g?

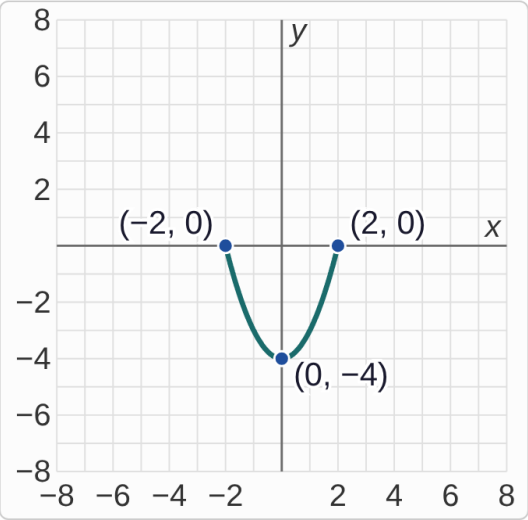
- **A.** (0, −3), (2, −1), (4, −3)
- **B.** (−7, −3), (−5, −1), (−3, −3)
- **C.** (−7, 0), (−5, 2), (−3, 0)
- **D.** (−4, −3), (−2, −1), (0, −3)

Answer and reasoning

Left 2, then down 5: *(−2,2)* goes to *(−4,−3)*, *(0,4)* goes to *(−2,−1)*, *(2,2)* goes to *(0,−3)*. The tent keeps its shape, with its peak now at *(−2,−1)*. The list starting at *(0,−3)* goes right, the one starting at *(−7,−3)* uses 5 for both shifts, and the one with *(−5,2)* swaps the two numbers. ***(−4,−3), (−2,−1), (0,−3)*.**

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Example 8: Track 4



Plotted data: The graph of f on a grid from −8 to 8 in both directions. It is a piece of the parabola y = x squared − 4, drawn only from x = −2 to x = 2. It starts at (−2, 0), goes down to its lowest point (0, −4), and comes back up to (2, 0). All three points are labeled.

$$g(x)=f(x-4)+1$$

Question

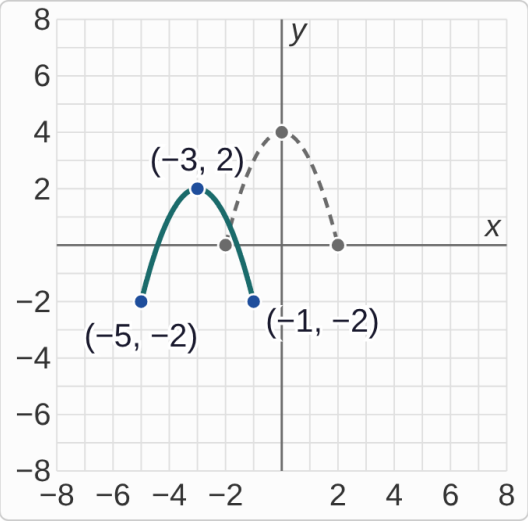
The domain of f is [−2, 2]. What is the domain of g?

- **A.** [−6, −2]
- **B.** [−2, 2]
- **C.** [2, 6]
- **D.** [−1, 3]

Answer and reasoning

The domain is the set of *x*-values, so only the sideways shift changes it. Right 4 moves *[−2,2]* to *[2,6]*. The *+1* moves the graph up, which changes the range, not the domain. The interval *[−6,−2]* goes left, and *[−1,3]* uses the 1 instead of the 4. ***[2,6]*.**

Example 9: Match 1



Plotted data: Grid from −8 to 8. Dashed: the graph of f, the arch y = 4 − x squared from (−2, 0) up to (0, 4) and down to (2, 0). Solid: the graph of g, the same arch shape from (−5, −2) up to its top point (−3, 2) and down to (−1, −2). The three points of g are labeled.

Question

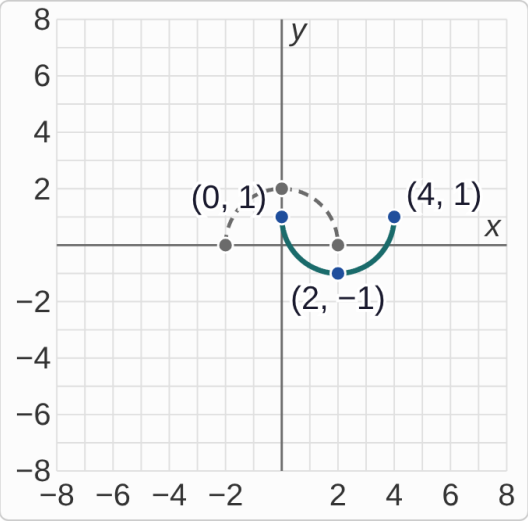
Which formula gives g?

- **A.** $g(x)=f(x-3)-2$
- **B.** $g(x)=f(x+3)+2$
- **C.** $g(x)=f(x+2)-3$
- **D.** $g(x)=f(x+3)-2$

Answer and reasoning

Compare the top points: *(0,4)* moved to *(−3,2)*, so the graph went left 3 and down 2. The arch is still right side up, so there is no reflection. Left 3 is *x+3* inside, and down 2 is *−2* outside. The *x−3* version goes right, the *+2* version goes up, and *f(x+2)−3* swaps the 3 and the 2. ***g(x)=f(x+3)−2*.**

Example 10: Match 2



Plotted data: Grid from −8 to 8. Dashed: the graph of f, the upper semicircle from (−2, 0) up to (0, 2) and down to (2, 0). Solid: the graph of g, a bowl-shaped half circle that starts at (0, 1), goes down to its lowest point (2, −1), and comes back up to (4, 1). The three points of g are labeled.

Question

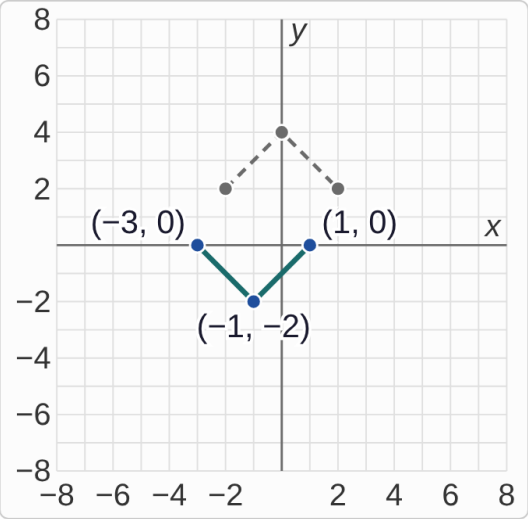
Which formula gives g?

- **A.** $g(x)=-f(x-2)+1$
- **B.** $g(x)=f(x-2)-1$
- **C.** $g(x)=-f(x+2)+1$
- **D.** $g(x)=-f(x-2)-1$

Answer and reasoning

The half circle is upside down, so there is a reflection: a minus in front. Its center moved from $x=0$ to $x=2$, right 2, so $x-2$ inside. After the reflection the middle point would be at $(2,-2)$, and it is at $(2,-1)$, so up 1. Check an endpoint: $g(0)=-f(-2)+1=-0+1=1$. Without the minus, the graph would be right side up; $x+2$ goes left; and -1 would put the endpoints below the axis. ***g(x)=−f(x−2)+1*.**

Example 11: Match 3



Plotted data: Grid from −8 to 8. Dashed: the graph of f, the tent from (−2, 2) up to (0, 4) and down to (2, 2). Solid: the graph of g, a V shape from (−3, 0) down to its lowest point (−1, −2) and back up to (1, 0). The three points of g are labeled.

Question

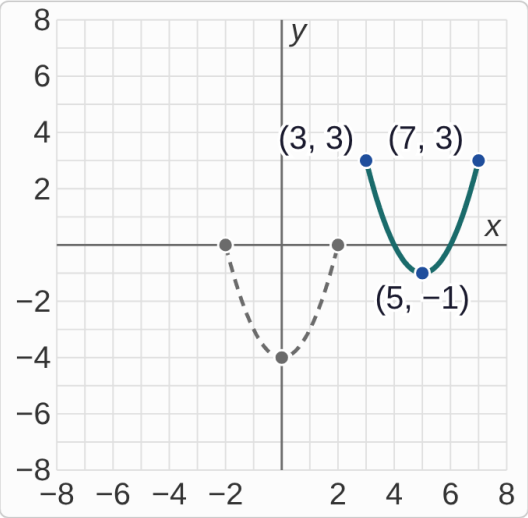
Which formula gives g?

- **A.** $g(x)=-f(x-1)+2$
- **B.** $g(x)=f(x+1)-2$
- **C.** $g(x)=-f(x+1)+2$
- **D.** $g(x)=-f(x+1)-2$

Answer and reasoning

The tent is upside down, so a minus in front. The peak moved from $x=0$ to $x=-1$, left 1, so $x+1$. Reflected, the peak would be at $(-1,-4)$; it is at $(-1,-2)$, so up 2. Check: $g(-3)=-f(-2)+2=-2+2=0$. The formula $f(x+1)-2$ has no reflection, so its tent would point up. The formula $-f(x+1)-2$ would put the low point at -6, and $-f(x-1)+2$ goes right. ***g(x)=−f(x+1)+2*.**

Example 12: Match 4



Plotted data: Grid from −8 to 8. Dashed: the graph of f, the bowl y = x squared − 4 from (−2, 0) down to (0, −4) and up to (2, 0). Solid: the graph of g, the same bowl shape from (3, 3) down to its lowest point (5, −1) and up to (7, 3). The three points of g are labeled.

Question

Which formula gives g?

- **A.** $g(x)=f(x+5)+3$
- **B.** $g(x)=f(x-5)+3$
- **C.** $g(x)=f(x-3)+5$
- **D.** $g(x)=-f(x-5)+3$

Answer and reasoning

The bowl still opens up, so no reflection. Its low point moved from $(0,-4)$ to $(5,-1)$: right 5 and up 3. Right 5 is $x-5$. The $x+5$ version goes left, $f(x-3)+5$ swaps the 5 and the 3, and the minus in front would turn the bowl into an arch. ***g(x)=f(x−5)+3*.**

Example 13: Slip 1

$$g(x)=f(x+3)$$

Question

A student says the graph of g is the graph of f shifted right 3, because the sign is plus. What is the mistake, if any?

A. There is no mistake.
B. It is a shift up 3, because it adds 3.
C. A plus inside shifts left, so it is left 3.
D. It is a reflection across the y-axis, then a shift right.

Answer and reasoning

Inside the parentheses, the shift goes the opposite way from the sign. The point that was at $x=0$ now appears at $x=-3$, since $g(-3)=f(0)$. **The graph shifts left 3.**

Example 14: Slip 2

$$g(x)=f(x-4)-1$$

Question

A student says the point (0, 4) on f moves to (4, 3) on g. What is the mistake, if any?

A. There is no mistake.
B. It should be (−4, 3), since x − 4 moves left.
C. It should be (4, 5), since − 1 moves up.
D. It should be (−4, 5), moving left and up.

Answer and reasoning

$x-4$ moves right 4, and -1 moves down 1: $(0,4)$ goes to $(4,3)$. Check: $g(4)=f(4-4)-1=f(0)-1=4-1=3$. **There is no mistake.**

Example 15: Slip 3

$$g(x)=-f(x-1)+2$$

Question

A student described g as: shift right 1, shift up 2, then reflect across the x-axis. What is the mistake, if any?

A. There is no mistake; the order does not matter.
B. It should shift left 1, not right 1.
C. The reflection should be across the y-axis, not the x-axis.
D. The reflection has to come before the shift up 2.

Answer and reasoning

The minus multiplies $f(x-1)$, not the $+2$. So reflect first, then shift up 2. The student's order gives $-(f(x-1)+2)=-f(x-1)-2$, which ends up 4 lower. **Right 1, reflect across the x-axis, then up 2.**

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Example 16: Slip 4

$$g(x)=-f(x)$$

Question

A student says the graph of g is the graph of f reflected across the y-axis. What is the mistake, if any?

A. There is no mistake.
B. A minus in front reflects across the x-axis instead.
C. It is a shift down, because of the minus sign out front.
D. It is a reflection across the line y = x instead.

Answer and reasoning

$-f(x)$ keeps every x and changes the sign of every y: (x,y) goes to $(x,-y)$. That flips the graph across the x-axis. A reflection across the y-axis would be $f(-x)$, which changes the sign of x. For CheckIt's three shapes, which are symmetric, that would not change the graph at all. **It reflects across the x-axis.**

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Example 17: Slip 5

$$g(x)=-f(x+2)-3$$

Question

A student says the point (0, 4) on f moves to (−2, 1) on g. What is the mistake, if any?

A. It should be (2, 1), since x + 2 moves right.
B. There is no mistake.
C. The reflection was skipped; it should be (−2, −7).
D. It should be (−2, −1), reflecting after the shift down.

Answer and reasoning

Left 2 gives $(-2,4)$. Reflecting across the x-axis gives $(-2,-4)$. Down 3 gives $(-2,-7)$. The student did the shifts but not the reflection. Check: $g(-2)=-f(0)-3=-4-3=-7$. **It moves to $(-2,-7)$.**